

Thermodynamic State Vector Discrepancy Absolute Difference Math Function: Sprint 069 Implementation Report

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Web of Life

Sprint 069 – Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Computational ecosystem models require stringent enforcement of thermodynamic laws to maintain physical validity. Sprint 069 introduces the pure helper function `computeAbsoluteStockDelta(actual, expected)` within `src/thermodynamics/state_validator.ts`. This module formalizes thermodynamic state vector discrepancy evaluation by calculating exact absolute differences across elemental and energetic stock keys. Framed through non-equilibrium thermodynamics and systems ecology, this operator supports side-effect-free monadic validation of mass-energy conservation (First Law) and entropy tracking (Second Law).

1 Introduction and Systems Ecology Context

The *Web of Life* simulation engine models complex biogeochemical cycles bounded by strict thermodynamic constraints. To prevent numerical drift and uphold ecological realism, state vectors representing matter (C, H, O, N, P, H_2O) and energy (J) must be continually evaluated against conservation baselines. Sprint 069 delivers a standardized, pure discrepancy operator designed for seamless integration into functional monad pipelines without violating immutability.

2 Thermodynamic Stock Transfer Equations

Let $\vec{S}_{act}$ be the actual state vector containing elemental stocks, and let $\vec{S}_{exp}$ be the expected baseline vector. For each key $k \in K$, where $K = \text{Keys}(\vec{S}_{act}) \cup \text{Keys}(\vec{S}_{exp})$, the absolute stock delta Δ_k is computed via:

$$\Delta_k = |S_{act,k} - S_{exp,k}| \quad (1)$$

Where missing keys default to the additive identity (0):

$$S_{act,k} = \begin{cases} val & \text{if } k \in \vec{S}_{act} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

3 Architecture and Implementation

The discrepancy calculation is encapsulated as a pure monad utility method preserving immutability across state transformations:

```

import { StateVector } from './state_vector';

export function computeAbsoluteStockDelta(
  actual: StateVector | Record<string, number>,
  expected: StateVector | Record<string, number>
): Record<string, number> {
  const result: Record<string, number> = {};

  const actualStocks = (actual instanceof StateVector) ? actual.getStocks() : actual;
  const expectedStocks = (expected instanceof StateVector) ? expected.getStocks() : expected;

  const allKeys = new Set([...Object.keys(actualStocks), ...Object.keys(expectedStocks)]);

  for (const key of allKeys) {
    const actVal = actualStocks[key] ?? 0;
    const expVal = expectedStocks[key] ?? 0;
    result[key] = Math.abs(actVal - expVal);
  }

  return result;
}

```

4 Conservation Tracking & Compliance

- **Mass Conservation (C, H, O, N, P, H_2O):** Quantifies mass divergence against closed-loop stoichiometry.
- **Energy Conservation (J):** Tracks thermodynamic dissipation or unaccounted heat sources.
- **Monad State Integrity:** Guarantees zero mutation of input state vectors during discrepancy evaluation.

5 Conclusion

Sprint 069 establishes a robust mathematical foundation for thermodynamic validation within the *Web of Life* simulation engine, ensuring high fidelity in biophysical and ecological modeling.